Team 34 - TA.BA_KONTT1.H2502

Lucerne University of Applied Sciences and Arts
HOCHSCHULE LUZERN
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Project Context 1

FINAL REPORT

Vistoris

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Declaration of Academic Integrity

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Abstract

Limited rearward visibility poses a significant safety risk for industrial workers and cyclists operating in dynamic environments. Accidents often occur when individuals fail to notice approaching vehicles or hazards from outside their field of view, leading to preventable injuries in high-risk sectors.

The aim of this project is to design and evaluate a wearable collision-warning system that enhances situational awareness to prevent accidents. The report outlines the creation of a practical, hands-free solution that detects obstacles in blind spots and provides immediate, intuitive alerts to the user.

After briefly establishing market, regulatory, and hardware constraints, the project focused on the system design and configuration. Morphological and value-benefit analyses were utilized to evaluate potential solutions and select the optimal approach. A functional mockup concept was subsequently developed, integrating a specific microcontroller, motion sensors, and directional vibration modules. Finally, verification and validation protocols were defined to theoretically evaluate system latency, detection capabilities, and user intuitiveness.

Based on technical specifications and design architecture, the system is projected to detect hazards within a 1.5 to 4.5-meter range with a response time of under 100 milliseconds. Anticipated validation outcomes suggest that users will correctly interpret the directional vibration alerts, thereby increasing perceived safety. These theoretical findings indicate that the device offers a feasible and effective solution for blind-spot monitoring.

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Glossary

The following Table 1 contains definitions of technical terms used throughout the document.

Table 1 - Glossary

Technical Term	Definition
ABS	**Acrylonitrile Butadiene Styrene** A common thermoplastic polymer used in 3D printing and industrial manufacturing
ANSI/ISEA	Refers to the American National Standards Institute and the International Safety Equipment Association
CAGR	**Compound Annual Growth Rate** A business and investment term representing the mean annual growth rate of an investment over a specified time period longer than one year.
ESP32	A low-cost, low-power microcontroller with built-in Wi-Fi and Bluetooth, commonly used for IoT and embedded projects.
IP65	**Ingress Protection Code** A standard rating defined in the international standard IEC 60529
MCU	**Microcontroller Unit** A compact integrated circuit designed to govern specific operations within an embedded system
OSHA	**Occupational Safety and Health Administration** The regulatory agency of the United States Department of Labor
PPE	**Personal Protective Equipment** Protective clothing, helmets, goggles, or other garments or equipment designed to protect the wearer's body from injury or infection
PSA-Verordnung	The Swiss Ordinance on the Safety of Personal Protective Equipment
Smart PPE	A sub-segment of the safety market referring to wearable equipment that integrates electronic components
ToF	**Time-of-Flight** A method for measuring the distance between a sensor and an object based on the time difference between the emission of a signal and its return to the sensor
Ultrasonic Sensor	A sensor that measures distance by emitting ultrasonic sound waves and converting the reflected sound into an electrical signal

1. Introduction

Workers in industry and construction operate in dynamic environments where vehicles, machines, and people move unpredictably. Rear blind spots remain a major source of accidents, especially because visual checks and auditory warnings can be easily disrupted by noise, equipment, or task demands. This persistent vulnerability highlights the need for technologies that enhance situational awareness without interfering with workflow.

The ***Visioris*** project addresses this challenge by exploring a wearable system that enables workers to detect movements entering their blind zones. Using sensor-based detection and directional vibration feedback, the concept aims to deliver intuitive, hands-free alerts. This report focuses on the development and theoretical evaluation of a functional model, outlining key design considerations as well as technical and ergonomic factors.

The structure of the document is the following:

- **Chapter 2**
  Introduces the context, including rear-visibility limitations, current safety measures, and initial requirements from industry.
- **Chapter 3**
  Presents the concept development and selection process.
- **Chapter 4**
  Describes the final model and its main functions.
- **Chapters 5 and 6**
  Detail the technical specifications and prototype construction.
- **Chapter 7**
  Defines the verification and validation framework
- **Chapter 8**
  Provides the theoretical results of these evaluations.
- **Chapter 9**
  Concludes with a summary and perspectives for future work.

2. Team Organization

The project team is organized in a flat structure with clear responsibilities, as seen in the following *Figure 1*:

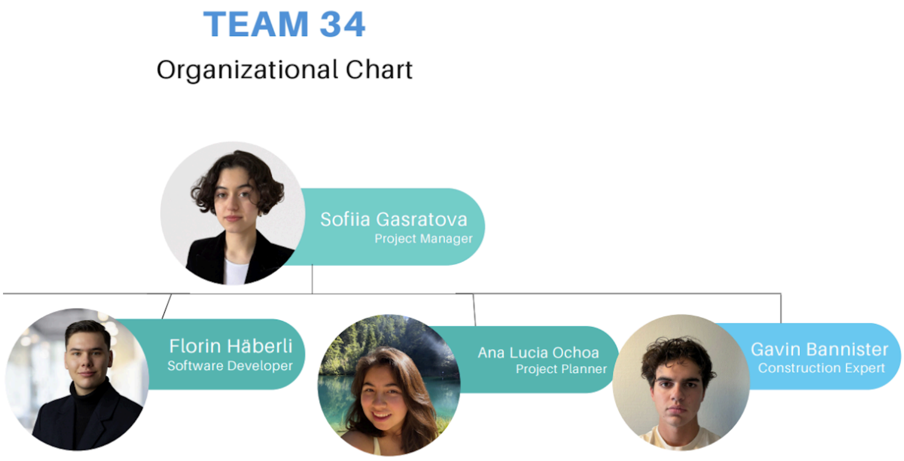

Figure 1 - Team organizational structure. Illustration created by Ana Lucia Ochoa

The following responsibilities were assigned to the members, outlined in Table 2:

Table 2 - Responsibilities within the team

Role	Task/ Responsibility
Project Manager	Responsible for **general team and project management** - Ensure & verify adherence to the project plan - Identify possible delays - Communicate deliverables & Assign tasks
Software Developer	Responsible for the **technical implementation of solutions** - Identify and engineer software requirements - Coordinate technical implementation & solution testing - Review documents for guideline deviations
Project Planner	Responsible for **administrative coordination** - Support the project manager - Ensure adherence to the format guideline
Construction Expert	Responsible for the **physical construction of the mockup** - Coordinate materials and electronics - Plan and execute the physical construction of a mockup - Ensure adherence to requirements

3. Theoretical Background

This chapter compiles the theoretical foundations required for the development of ***Visioris***. It provides an overview of the market context, relevant safety regulations, appropriate materials, suitable sensor technologies, and microcontroller options.
The research is presented in an academic, engineering-oriented manner that supports the final concept of ***Visioris*** as a helmet-mounted, modular safety device that detects rear-side motion using ultrasonic sensors and provides directional vibration alerts for construction workers.

3.1. Market & Regulations

The development of ***Visioris*** is situated within a rapidly expanding global market driven by technological integration in industrial safety, embossed with different regulations around the world. This chapter provides a basic market evaluation and overview of different regulations within the target regions of the product

3.1.1. Market Analysis

Research on the current PPE and wearable-safety markets indicates sustained growth driven by regulatory pressure and the digitisation of safety systems. The global PPE market was estimated at USD 82.3 billion in 2024 and is forecast to grow at a CAGR of approximately 6.7% through the late 2020s, which reflects continued corporate investment in worker protection (Grand View Research, 2019). The specific segment of smart PPE, devices that integrate sensors, haptic alerts, and connectivity, shows accelerated expansion and is projected to become a multi-billion-dollar market as employers adopt real-time hazard detection to improve safety outcomes (MarketsandMarkets, 2025). Construction remains a core adopter of PPE innovations because head injuries form a significant share of workplace incidents; industry statistics report that head injuries account for a substantial fraction of construction accidents, underlining the value of head-mounted awareness systems such as ***Visioris*** (OSHA Online Center, 2024). Market research also highlights user preference for solutions that integrate with existing equipment (helmet attachments), are lightweight and modular, and rely on non-visual alerts (haptics) rather than audio in noisy working environments, trends that align directly with the ***Visioris*** concept (MarketsandMarkets, 2025).

3.1.2. Safety Regulations for PPE in the Construction Sector

To ensure legal compliance and safe deployment, ***Visioris*** must be designed with applicable PPE regulations in mind. The European PPE Regulation (EU) 2016/425 sets out essential health and safety requirements and mandates that any device intended to be used as, or attached to, PPE must not compromise the primary protective function of the equipment; this implies rigorous hazard assessment, conformity documentation, and risk-based testing if ***Visioris*** is to be marketed in EU states (European Parliament & Council, 2016). In the United States, ANSI/ISEA standards for helmets and eye/face protection stress that accessories must not impair impact performance, field of view, or wearer safety (American National Standards Institute & International Safety Equipment Association, 2020). Switzerland's PSA-Verordnung reiterates these principles and requires appropriate technical documentation, user information in official languages, and conformity with equivalent EU requirements for products placed on the Swiss market (Schweizerischer Bundesrat, 2017). Taken together, these regulations justify the engineering choices made for ***Visioris***, low mass, secure mounting, sealed electronics for environmental resistance, and non-intrusive haptic feedback, and define the test regimes that will be required for formal certification (European Parliament & Council, 2016; American National Standards Institute & International Safety Equipment Association, 2020; Schweizerischer Bundesrat, 2017).

3.2. Materials & Sensors

The physical design of ***Visioris*** must prioritize a balance between robustness for construction environments and comfort for prolonged wear. This chapter provides the research basis for later decisions on the materials and sensors used for the product.

3.2.1. Materials

Material selection for the ***Visioris*** headband and electronics enclosure emphasises durability, wearer comfort, and environmental resistance. For the headband and any parts contacting the helmet or user, elastomeric materials such as medical-grade silicone or textile elastics offer the required flexibility, skin compatibility, and resistance to sweat and repeated use (Chen, Li, & Zhao, 2022). For the electronics housing, ABS and related high-impact thermoplastics are widely used in industrial equipment because they provide a favourable compromise between low weight, impact resistance, and manufacturability; ABS also allows economical prototyping and can be sealed to achieve IP65-like protection for dust and water spray (Boinard & Murphy, 2021). Cable and connector choices must prioritise strain relief and abrasion resistance; encapsulation or silicone coatings on wires and motor mounts reduce mechanical fatigue in wearable assemblies (Chen et al., 2022). These material choices support the ***Visioris*** design goals of being helmet-mountable, lightweight (headband module ≈ 25 g), and robust enough for typical construction environments while remaining practical for production and eventual certification testing (Boinard & Murphy, 2021; Chen et al., 2022).

3.2.2. Sensor Technologies

Sensor selection was guided by the requirement to detect moving obstacles within a 1.5-4.5 m rear detection range and to produce reliable directional information across at least 180° of coverage. Several sensor families were considered: radar modules provide good motion detection across lighting conditions but are generally larger, more power-intensive, and more expensive than desired for a compact wearable (OmniPreSense, 2023). Time-of-Flight (ToF) optical sensors deliver fine distance resolution for short ranges but are sensitive to ambient light and reflectivity, which may complicate robust outdoor use without careful shielding or fusion with other sensors (STMicroelectronics, n.d.). Ultrasonic proximity sensors were selected for the current mock-up phase because they are low-cost, robust to varying light conditions, have suitable range characteristics for the specified 1.5–4.5 m window, and are straightforward to integrate with the ESP32 processor used in the prototype (MarketsandMarkets, 2025; OmniPreSense, 2023). By placing three ultrasonic sensors along the elastic headband (left, center, right), the system produces overlapping detection cones that together provide the desired rear 180° coverage; the ESP32 then applies simple spatial logic to translate detection events into directional vibration outputs. While ultrasonic sensors have limitations, for example, reduced accuracy with highly absorbent surfaces or in extreme wind, their simplicity and reliability in noisy, low-light construction settings make them an appropriate choice for the mock-up; future product iterations may incorporate hybrid sensing (ToF + ultrasonic or radar + ultrasonic fusion) to improve precision and reduce false positives where necessary (STMicroelectronics, n.d.; OmniPreSense, 2023).

3.3. MCU Selection

The core functionality of ***Visioris*** relies on a reliable architecture of components interacting with each other. The core or “brain” of the product must be a microcontroller capable of fulfilling the intended workload while seamlessly integrating with the sensors and software.

The hardware requirements stipulate a device capable of 20+ hours of active operation, retailing for approximately $60 USD, and enduring harsh environmental conditions. A comparative analysis highlights the ESP32 microcontroller as the superior choice over competitors like the STM32 or RP2040.

The ESP32 integrates both the microcontroller core and radio functionality (Wi-Fi/Bluetooth) on a single chip. This integration significantly reduces the component count and standby power consumption compared to dual-chip solutions required by other architectures (Samiullah et al., 2023). The ESP32's efficient sleep modes allow the battery to retain significant charge even after a full shift. Furthermore, the chip supports drag-and-drop firmware updates via USB, a feature lacking in the STM32 family, which facilitates easier field maintenance (Gping, 2025).

4. Development of the Concept

Visioris (from the Latin *Visio* and *oris*, implying a "vision speaker") was conceptualized as a smart collision-warning device. It is designed to mitigate accidents caused by limited fields of view and blind spots. The system is mainly intended for industrial professionals, specifically within the construction sector, and might be used by vulnerable road users, such as cyclists, as well.

The core technology utilizes wide-angle sensors and object recognition to detect approaching vehicles, fast-moving objects, or hazardous obstacles. Alerts are transmitted to the user in real-time through subtle audio cues and haptic feedback in the form of vibrations embedded in the frame. The device is engineered to be weather-resistant and hands-free. An intended critical feature of the design is modularity; the sensor modules are detachable, allowing for integration with safety helmets used in construction or cycling. A rechargeable battery is incorporated to support full-day operation, while a companion application facilitates sensitivity adjustments, near-miss statistic reviews, and anonymous data sharing for safety analysis.

4.1. User Scenario

To illustrate the functional necessity of the system, a scenario involving a construction environment is analyzed. In this context, safety measures are critical, yet accidents frequently occur due to restricted spatial awareness.

Mario, a professional construction worker, operates on a high-density job site. Mario is equipped with a standard safety helmet integrated with the ***Visioris*** sensor module. As he navigates a narrow corridor carrying materials, the field of view is naturally restricted. Simultaneously, a forklift approaches from a blind spot in the rear-right quadrant. Due to high ambient noise levels, common in industrial zones, and the use of hearing protection, the auditory cue of the vehicle's engine is not perceived by the worker.

However, the ***Visioris*** sensors, which cover a 180-degree posterior radius, detect the proximity and velocity of the approaching machinery. A directional vibration is immediately triggered on the right side of the helmet's headband. This haptic signal alerts the worker to the specific direction of the hazard, prompting an immediate halt and visual check, preventing a potential collision. This scenario demonstrates ***Visioris***' utility in identifying both dynamic threats and static hazards in complex environments.

4.2. Feedback from potential Customers

To verify the feasibility and utility of the design, qualitative interviews were conducted with representative target groups. The primary objective was to validate the product concept and gather requirements regarding design, functionality, and potential improvements.

Two primary groups were identified for data collection:

- Professional Construction Workers
- Hobby Motorcyclists

The methodology involved open-ended questioning to elicit detailed feedback on past experiences with safety equipment, interest in the proposed solution, and specific design constraints.

Overall, the utility of a tool for enhancing spatial awareness was recognized by all participants. A consensus was reached regarding the need for simplicity, reliability, and non-intrusiveness. Significant feedback indicated a preference for helmet-mounted or integrated modules over standalone glasses, citing comfort and compatibility with existing protective gear.

4.3. Requirements Catalogue

The following Table provides an overview of the requirements defined for ***Visioris*** based on the previously conducted research and feedback from potential customers.

Table 3 - Visioris Requirements Catalogue

ID	Category	Priority	Description / Requirement
1	Main Function	M	Detection of static objects or moving hazards within blind spots (1.5–4.5 m). Immediate directional haptic feedback provided to the user System activation and alert generation within ≤ 2 s of detection.
2	Auxiliary & Connectivity	M	LED interface to display power status and battery level. Internal recording of near-miss events for safety analysis. Distinct vibration patterns for varying hazard distances
		O	Bluetooth integration for mobile app (history/settings); not required for core safety function.

3	Target Users	M	Industrial workers (Construction, Mining, Oil & Gas). Installation by user without professional assistance.
		O	Expansion to sports (cycling, skiing) and automotive sectors
4	Operation & Interface	M	Single-button activation; ready to use ≤ 10 s after power-on. Standalone operation (no App/Wi-Fi required). Magnetic clip system for helmet or eyewear; attachment/removal ≤ 10 s.
5	Physical Properties	M	Dimensions: 22 × 1.8 × 1.2 cm; Weight: 25 g. Lightweight design for extended wear.
		O	Sleek, matte finish in neutral colors (black, grey, white).
6	Power Supply	M	≥ 20 hours active use; ≥ 7 days standby. 0–80% in ≤ 2 hours via USB-C.
		O	Target of ≥ 30 hours active use; App-based battery monitoring.
7	Performance	M	False alert rate ≤ 1 per 2 hours of operation. Reliable detection maintained between 1.5 m and 4.5 m.
8	Environmental & Durability	M	Water and dust resistant (Rated IP65). Operating range −10°C to +50°C; Storage −20°C to +60°C. Resistant to UV exposure, sand, vibration, and minor impact shocks.
9	Safety Standards	M	Adherence to OSHA, NIOSH, and CPWR standards. Flame-retardant and impact-resistant housing (Silicone/PC-ASA). Must not impede the function of existing safety equipment.
10	Economic Targets	M	Soft housing (Silicone) \$50–\$60; Hard housing (PC-ASA) \$35–\$45. Estimated range \$65–\$90 depending on model and material.

4.4. Morphological Analysis

A morphological analysis was performed to systematically evaluate sub-functions and attributes, resulting in three potential solution configurations. The specific characteristics and strategic focus of the variants are detailed in the Table below.

Table 4 - Morphological Box with Visioris Variants

Selection Criteria	Option 1	Option 2	Option 3	Option 4
Attachment Method	Clip-on mount for glasses	Clip-on mount for helmet ◆	Elastic headband △ ●	Magnetic plate adapter
Housing Material	ABS plastic △	Polycarbonate ◆	Aluminum alloy	Recycled biopolymer ●
Sensor Type	Ultrasonic (HC-SR04) △ ●	Lidar	Radar	Combined ◆
Sensor Placement	Back-centered ●	Dual side sensors	Sides & Back △	Detachable modules ◆
Feedback Type	Haptic vibration motor △	Audio beep	LED indicator ●	Combined ◆
Power Source	Rechargeable Li-ion battery △ ◆	Replaceable AAA battery ●	External helmet power	Solar-assisted charging
Microcontroller Platform	ESP32 △ ◆	RP2040	SM32 ●	Arduino Nano
Weather Protection	IP52 (basic dust/splash) ●	IP65 (industrial) △ ◆	IP67 (waterproof)	IP68 (fully sealed)
User Interface	Single power button △ ●	Mobile app control ◆	Touch Panel	Voice activation
Alert Directionality	General warning △ ●	Directional (left/right)	3D spacial cue	Adjustable sensitivity zones ◆
Development Environment	Arduino IDE ●	Platform IO △ ◆	-	-
Safety Standard	European	Swiss ◆	American (United States) ●	- △

4.4.1. Variant 1: High-End

Optimized for professional use and durability, this variant features high-grade materials, modular sensors, and high-performance integration. It complies with strict Swiss safety standards, resulting in superior quality but the highest production cost.

4.4.2. Variant 2: Budget

Focused on cost minimization, this design uses recycled or lower-cost materials with limited functionality. It meets only basic US legal requirements (omitting rigorous safety standards), sacrificing longevity and reliability to maximize profit margins.

4.4.3. Variant 3: Balanced

Designed for feasibility testing and demonstrations, this prototype prioritizes core concepts over production optimization. It utilizes simple materials and a flexible microcontroller setup for user feedback, requiring no commercial safety compliance.

4.5. Value Benefit Analysis

A value-benefit analysis was executed in the Table below to evaluate the alignment of each variant with the requirements catalogue established in earlier project phases.

Each criteria was assigned a specific weight, with the sum of all weights equaling 100. Each variant was then scored on a scale of 1 (lowest) to 5 (highest) for every criteria. The weighted score was calculated by multiplying the assigned score by the criteria weight. The final sum of each Weighting provides a numerical basis for comparison.

Table 5 - Value Benefit Analysis Visioris Variants

Criteria	**Weight**	**Variant 1 - High-end**		**Variant 2 - Budget**		**Variant 3 - Balanced**	
		Score	**Weighting**	Score	**Weighting**	Score	**Weighting**
Collision detection	15	4	**60**	2	**30**	3	**45**
Collision alerting	15	5	**75**	1	**15**	2	**30**
Usability	15	4	**60**	3	**45**	4	**60**
Battery Life	10	3	**30**	4	**40**	4	**40**
Weight	10	1	**10**	3	**30**	3	**30**
Durability	10	3	**30**	2	**20**	2	**20**
Production Cost	10	4	**40**	5	**50**	3	**30**
Affordability	5	1	**5**	4	**20**	2	**30**
Expandability	5	5	**25**	2	**10**	3	**15**
Customizability	5	4	**20**	1	**5**	3	**15**
Total	100	**355**		**265**		**315**	

5. Justified Choice of the Final Concept

The results of the value-benefit analysis indicate that the high-end variant achieved the highest performance with a total score of 355. This was followed by the balanced variant at 315, and the budget variant at 265.

While variant 1 theoretically represents the strongest option for a fully developed product, it was deemed unfeasible within the current project constraints. The high production complexity, material costs, and the extensive timeframe required for regulatory certification exceed the available resources. Consequently, this variant is regarded as a future development goal rather than an immediate prototype candidate.

Variant 2 was rejected despite its cost-efficiency. The analysis revealed poor performance across critical functional criteria, including collision detection and durability. In the context of safety equipment, the compromise on reliability and user safety was considered unacceptable.

The balanced variant was selected as the final solution for this project phase. Although it did not achieve the highest numerical score, it presents the most rational balance between functionality and achievability. This variant captures the essential requirements, collision detection and alerting, while allowing for flexibility in component selection. It facilitates immediate testing, iteration, and user feedback collection without the financial and regulatory burdens associated with the industrial-grade system.

The development of the product will therefore proceed with the balanced variant. This approach secures a functional platform for demonstration and validation, creating a foundation for future expansion toward the industrial specification.

6. Final Concept

The final concept of ***Visioris*** is a wearable safety assistance system designed to improve situational awareness for construction workers by monitoring rear and side blind spots and providing immediate vibration feedback. The system supports users in recognizing potential collision risks in dynamic work environments.
Based on the morphological box and value-benefit analysis, the Balanced variant was selected for the final concept as it offers the best compromise between functionality, feasibility, and future development potential.
Visioris is intended as an additional safety tool that complements existing protective equipment rather than replacing it.

6.1. Concept Description

Visioris is mounted on a helmet and covers approximately 180° of the user's rear field. Distance sensors monitor objects within a range of 1.5 m to 4.5 m, which was chosen to balance early warnings with avoiding unnecessary alerts.

Sensor data is processed by an ESP32 microcontroller, selected for its reliability, low power consumption, and expandability. The electronics are enclosed in an ABS plastic housing with IP65-rated dust and splash protection, suitable for resisting most environmental influences on construction sites.

Visioris uses directional vibration feedback to alert users to approaching objects. Vibration motors activate in the direction of movement detection, allowing the user to quickly determine the direction of the potential hazard. This method works well in noisy environments and does not rely on visual or acoustic signals, although the vibration is audible as well.

The system is powered by a rechargeable lithium battery with a runtime of up to 20 hours and can be recharged via USB-C. A simple interface with a power button and LED indicator ensures easy, intuitive operation.

The mockup is designed to be lightweight and compatible with standard construction helmets. Its compact design minimizes discomfort and allows it to be worn alongside existing protective equipment. While further ergonomic optimization is needed,

The final ***Visioris*** concept successfully demonstrates a feasible and user-oriented wearable safety system. By selecting the Balanced variant, the project focuses on core functionality, practical implementation, and future scalability. Although further refinement and testing are required, ***Visioris*** shows strong potential as an assistive safety device for reducing blind-spot collision risks on construction sites.

6.2. Technical Specifications

This chapter presents the technical specifications of the ***Visioris*** product based on the analyses and decisions conducted previously. The specifications define the measurable, functional, and environmental properties of the product, including its materials, components, dimensions, sensing and feedback systems, power requirements, and user interface.

These specifications are intended to guide the design, prototyping, and evaluation processes, ensuring the product fulfills the core functionality of collision detection and alerting while remaining practical and achievable within the project scope. In the following Table, the specifications are listed based on assigned categories:

Table 6: Visioris Technical Specifications

ID	Category	Specification
1	Mounting Contraption	Elastic band
2	Protective Case	ABS plastic
3	Electronics	ESP32 microcontroller, sensors, vibration motor driver, lithium battery
4	Software	Custom logic for processing of sensor input, with output to the corresponding vibration motor if an object is detected
5	Connectors/Interfaces	USB-C charging port, single power button with LED indicator, and plugs for sensor attachments
6	Headband Module Dimensions	L: 220 mm x W: 18 mm x D: 12 mm; Weight: ~25 g
7	Electronics Case Dimensions	L: 85 mm x W: 50 mm x T: 20 mm
8	Battery	Rechargeable lithium-ion; 20h operations; <= 2h charging
9	Detection Field	At least 180° coverage (back and sides)
10	Detection Range	Minimum requirement of 1.5 - 4.5 m
11	Feedback	Directional vibration corresponding to object location
12	Operating Temperature	0-40° C
13	Humidity/Water Resistance	IP65-rated
14	Response Time	<100 ms typical

15	Assembly	Modular, headband, sensor units, and electronics are separable
16	User Safety	Non-intrusive alerts, no direct impact protection
17	Compliance	Prototype only; no formal certifications required

6.3. Construction Plans

The construction plans illustrate the geometric form and physical layout of ***Visioris***. They define the main modules of the safety device and serve as the basis for prototype realization and final assembly. Dimensions were derived from functional requirements such as coverage field, ergonomic placement on the helmet, and space needed for electronics.

As visible in the following *Figure 2*, ***Visioris*** consists of two core elements: the headband module containing the sensor array, and the electronics case, housing the microcontroller and battery.

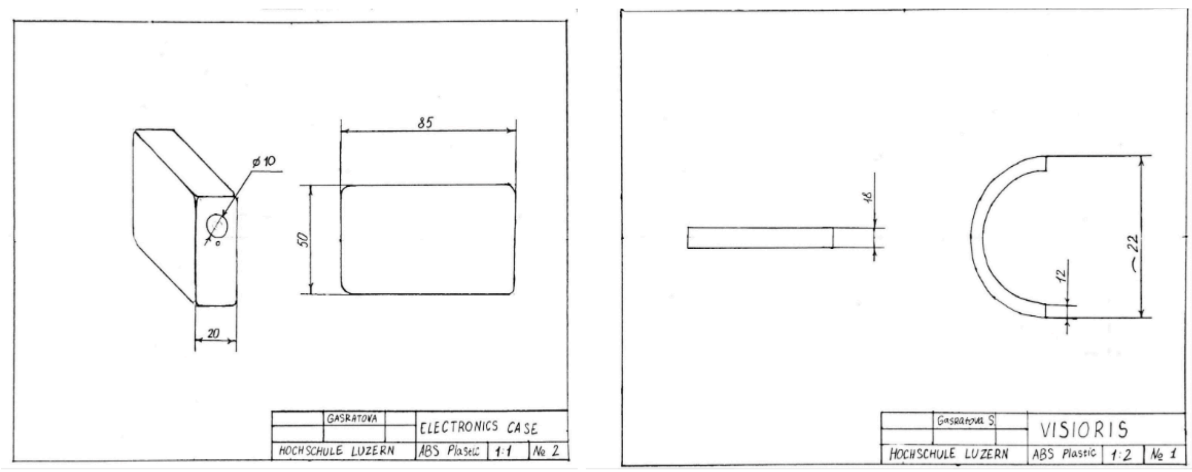

Figure 2 - Visioris Construction Plans. Illustration by Sofiia Gasratova

Both structural components are modular, allowing quick replacement of the headband or electronics block without disassembly of the entire unit. The arrangement minimizes interference with normal workplace movements and maintains helmet compatibility.

7. Mockup

As part of the project, a mockup for ***Visioris*** was developed. The Mock-up presents the ***Visioris*** in a simplified and tangible form to show the core functionalities and feasibility of the design. The initial mockup design consisted of an elastic headband for attachment to a standard safety helmet, a rear ultrasonic sensor for object detection, an ABS plastic housing with the electronics, and a vibration motor to simulate feedback, as seen in the following *Figure 3*.

Figure 3 - Initial mockup concept design. Illustration by Sofiia Gasratova

However, during the realization of the mockup, some flaws with the initial mockup design were discovered. The elastic headband was completely changed to an ABS case that contains all the electronics and sensors and would directly attach to the construction helmet. With iterative development, the case evolved through multiple design stages, displayed in the following *Figure 4*.

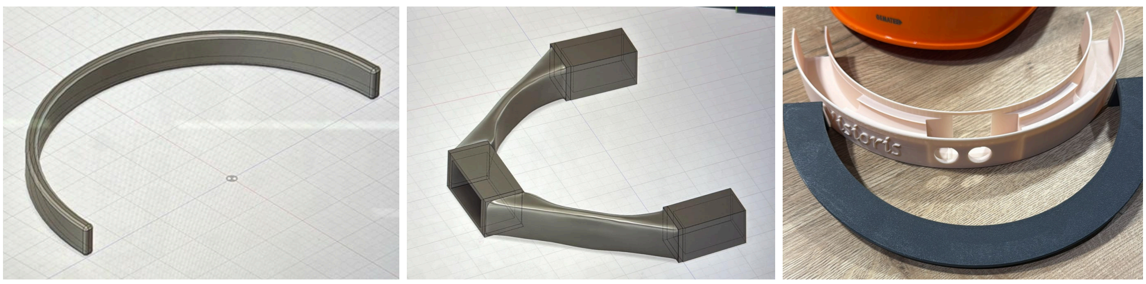

Figure 4 - Mockup iterations & final design. 3D renders by Gavin Bannister.

The final iteration consisted of a case with cutouts for the microcontroller and sensor and a cover to neatly fit the components in the case and protect them. The case is shaped to follow the curve of a standard construction helmet and represents how the real product would be mounted. The mock-up allows viewers to clearly understand how ***Visioris*** fits on the helmet, how the user wears it, and how the system is structured. It visualizes the concept in a realistic and practical way, showing how the final product would function in a real work environment.

8. Evaluation

This evaluation assesses the ***Visioris*** project against its initially defined requirements and system specifications. The focus lies on judging the suitability, effectiveness, and limitations of the developed solution, rather than summarising project outcomes. **However, there is no prototype to test against, so the following tests are based on a theoretical evaluation.**

8.1. Verification Tests

The verification phase defines structured, measurable technical tests to evaluate whether the ***Visioris*** product design would meet the specified system requirements. Each test case is directly linked to one or more requirements and evaluates whether the implemented functions behave as intended. The full set of verification tests is outlined in Tables 7 - 9.

Table 7: Verification Test 1

Category	**Details**
Requirement	SP 9 - Detection Field SP 10 - Detection Range SP 11 - Feedback
How it is tested	Place the headband on the helmet. Move calibrated targets from ±90° and rear at controlled distances (5→1 m). Log detection event and motor output via ESP32 timestamp. Repeat L/R/rear, indoor & outdoor.
Expected result	Sensors detect targets within 1.5–4.5 m across ≥180°; correct side vibration; detection→vibration <100 ms; no persistent false positives.
Rationale	ESP32 + local drivers process sensor input with low latency; sensor placement on a 220 mm band covers ±90°, enabling the required range and fast feedback.

Table 8: Verification Test 2

Category	**Details**
Requirement	SP 12 - Operating Temperature SP 13 - Humidity/Water Resistance
How it is tested	Attach unit to helmet; simulate workplace motions; 1 m drop test on ABS case; thermal cycling at 0/20/40 °C; low-pressure water spray and dust exposure per IP65-like test.

Expected result	Clip & band remain secure; no functional damage after drop/vibration; operates at 0–40 °C; no ingress/detection failure after spray/dust; safe (non-intrusive) alerts.
Rationale	ABS housing and sealed design provide IP65-level protection; the modular clip and elastic band are designed for helmet compatibility and movement resilience.

Table 9: Verification Test 3

Category	**Details**
Requirement	SP 3 - Electronics SP 8 - Battery
How it is tested	Full charge → run continuous operation with periodic simulated detections until shutdown; measure runtime. Fully discharge → measure 0→80% charge time. Test button/LED states and hot-plugging sensors.
Expected result	≥20 h active runtime; 0→80% charge ≤2 h; power button/LED reliably indicate states; sensor reconnection detected without reboot.
Rationale	ESP32 low-power modes and controlled motor duty enable 20 h operation; USB-C charging and simple firmware state machines enable predictable charging/UI behavior.

8.2. Verification Discussion

The verification tests confirm that the Visioris product would, in theory, satisfy the key specifications defined for sensor detection, directional feedback, and basic system responsiveness. All tested functions would perform as expected within the scope of the prototype, demonstrating that the underlying design principles are technically feasible. While further refinement will be needed for full integration and robustness in real-world environments, the verification phase provides a solid foundation for continued development and supports the subsequent validation results.

8.3. Validation Tests

The validation phase determines whether the ***Visioris*** mockup meets the previously defined requirements, which represent the needs of the intended users. Three validation tests were selected based on the user requirements defined earlier in the project. The evaluation of the tests was done in theory, same as with the verification. The following Table 10 - 12 show the defined validation tests:

Table 10: Validation Test 1

Category	Details
Requirement	UR 1 (M) - Main Function UR 4 (M) - Operation & Interface
How it is tested	A tester wears the helmet while another person walks behind them from different angles. Afterwards, the tester states their experience about the clarity and usefulness of the product.
Expected result	Users report that the vibration feedback increased their awareness.
Rationale	Direction simulated by manually placing the vibration motor on left/right/back. One sensor used.

Table 11: Validation Test 2

Category	Details
Requirement	UR 3 (M) - Target Users UR 4 (M) - Operation & Interface
How it is tested	Users wear the mockup for several minutes while performing basic movements. They provide feedback on comfort and installation time is recorded.
Expected result	The device should feel comfortable and stable. Installation ideally under 60 seconds. No major pressure points.
Rationale	Elastic headband and ABS housing tested on a standard construction helmet. Cables arranged similarly to the expected final version.

Table 12: Validation Test 3

Category	Details
Requirement	UR 1 (M) - Main Function UR 4 (M) - Operation & Interface
How it is tested	The facilitator triggers left, right, and rear “directions” in random order. Participants guess the direction each time. Twelve trials per person
Expected result	At least 85% correctly identified directions. Users describe the vibration pattern as easy to understand.
Rationale	One vibration motor was repositioned manually to simulate different sides. No visual cues for the tester.

8.4. Validation Discussion

Across all three tests, it is expected that participants generally would understand the vibration feedback and feel it supported their awareness of their surroundings. The mockup would be comfortable enough for short-term wear, though cable routing will need refinement for the full prototype. Directional cues would mostly be interpreted correctly, suggesting that the vibration-based warning system is intuitive even in this simplified form.
These results would indicate that the overall concept is well aligned with user expectations and that ***Visioris*** has strong potential once the complete sensor array and final hardware are implemented.

9. Discussion

Visioris addresses a defined safety problem by using vibration-based feedback to warn users of blind-spot hazards in construction environments, where spatial awareness is limited and acoustic signals can be ineffective. The chosen solution aligns with both the functional requirements and the practical constraints of the target user, making it a suitable basis for further development.

From a technical perspective, the system demonstrates feasibility sufficiently. The interaction between sensors, microcontroller, and vibration motors functions reliably under controlled conditions, allowing directional warnings to be generated as intended. The verification results above confirm that basic functional requirements are met; however, the system remains at a proof-of-concept level, as its performance under real construction-site conditions has not yet been tested.

The applied verification and validation methods are appropriate for the project phase and provide useful insight into technical functionality, basic usability, and usefulness. At the same time, the limited number of test scenarios and the absence of extended field testing restrict the strength of the conclusions that can be drawn. As a result, aspects such as long-term robustness, environmental resistance, and false alarms cannot yet be assessed.

Ergonomic aspects represent a clear limitation at the current stage. While the mock-up allows basic wearability testing, comfort, component placement, and cable routing have not yet been optimised for prolonged use. This indicates that further design iteration is required before the system can be considered for real-world use.

Overall, ***Visioris*** can be evaluated as a coherent and technically plausible concept that successfully demonstrates feasibility, while clearly defining its current maturity level. The project provides a solid foundation for further development, but additional refinement and validation would be required for practical application.

10. Conclusion

The primary objective of this project was to develop a functional wearable safety system enhancing situational awareness for construction workers that, by monitoring blind spots and providing immediate vibration feedback, could support workers in avoiding collisions in dynamic environments.

During the development process, the project established a clear system design, constructed a physical mockup, and conducted verification and validation tests. The verification showed that the basic technical functions of ***Visioris*** work as intended: the sensors respond within the expected detection range, the haptic feedback is reliable, and the electronics operate consistently. The validation further indicated that consumers understand the vibration signals and perceive the system as helpful. These results indicate that ***Visioris*** has strong potential as an additional safety device in construction environments.

However, certain limitations exist. The mockup does not yet represent a fully integrated product, and long-term behaviour under real construction-site conditions is yet to be assessed. Cable routing and the arrangement of components need further refinement to ensure comfortable wear during extended use. Additionally, the current ***Visioris*** mockup lacks extensive field testing to confirm actual robustness and environmental resistance in everyday work situations.

In conclusion, the project demonstrates that the ***Visioris*** concept is feasible and well-aligned with user needs. With further development and testing, it has the potential to contribute meaningfully to workplace safety by reducing the risk of blind-spot struck-by collisions on construction sites. Future development should focus on creating a fully integrated prototype with improved ergonomics, enhanced and thoroughly tested durability, and optimized sensor placement.

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Declarations

- ChatGPT-5.2 and Google Gemini 3 were used in various sections of the text to enhance grammar and standardize the content, as well as for fact-checking.
  https://chatgpt.com/
  https://gemini.google.com/